

7 A categorical dependent variable Logistic (logit) regression and related methods

7.1 Introduction

The y -variables in this book so far have been numerical, i.e. variables with a long range of numbers as possible values. In such cases, linear regression along with its extensions presented in [Chapter 6](#) most often works very well, even in the presence of mild violations of the regression assumptions. It is thus said that linear regression is a *robust* method. Quite often, however, research involves an y -variable with a few, categorical values. A special case is the binary variable; what we call a dummy. This chapter begins with analyzing a dummy y and finishes off with the analysis of a y with three unordered values/categories (see [section 7.6](#)).

Dummy dependent variables flourish in the social and economic sciences. Here are some examples:

- Some people choose to vote in elections, whereas others don't. Political scientists are interested in how independent variables such as age, gender and income affect this voting choice (yes = 1, no = 0).
- Some firms go bankrupt within a few years of startup, whereas others don't. Economists are interested in how independent variables such as type of funding, type of organizational structure and gender of CEO affect going bankrupt or not (yes = 1, no = 0).
- Some are employed full time, whereas others aren't. Sociologists are interested in how independent variables like education, working experience and family obligations affect this employment "choice" (yes = 1, no = 0).¹
- Some try to commit suicide, whereas most others (thankfully) don't. Psychologists are interested in how independent variables like type of upbringing, personality traits and self-esteem affect this "choice" (yes = 1, no = 0).

And the list goes on ... In short, we are talking about dependent variables that might be thought of as the presence (1) or no-presence (0) of something. Or, in the framework of questionnaires, a question that directly or indirectly may be answered with yes (1) or no (0).²

It has been customary for years to use logistic (often called logit) regression for analyzing a binary y . But of late it has become more popular to use the plain vanilla linear regression model even when y is binary. For this reason, this chapter starts with the so-called linear probability model (LPM), i.e. the regression model we have seen so far on a binary y ([section 7.2](#)). [Section 7.3](#) then uses the standard logistic (logit) regression model on the same example data as in [section 7.2](#). [Section 7.4](#) takes up the issue of interpreting and presenting the results of the logistic regression model, which is no small feat. Interaction effects, non-linearity and model fit are discussed in [section 7.5](#). [Section 7.6](#) presents a regression model for an extension in which y is a choice between three alternatives, and [section 7.7](#) addresses some further issues and points to further reading.

7.2 The linear probability model (LPM)

Scenario. A new fitness center is under construction in the vicinity of a university college. A random sample of students are in a questionnaire asked about how they would prefer to pay for exercising in this center. The options are (i) a fixed monthly fee with no restrictions on number of times exercising or (ii) paying a small fee per time for each exercise. (Both alternatives included cost information and were set up in such a way that exercising for two times per week amounted to the same monthly cost.) The frequency distribution and descriptive statistics for this variable (fixed_pm; yes = 1, no = 0) appear in [Stata-output 7.1](#). The data are called pay_pref_fit.

In the upper part of the output we see that about 73% of the students prefer to pay a fixed amount per month. Conversely, 27% prefer paying per time. The lower part tells us the same thing, but expressed as a probability or chance. The probability (chance) that a random student prefers a fixed price payment structure is about 0.73. Note that the mean of a 0/1 variable always is the *proportion* with the value of 1 on the variable.

Research question. The research question involves how two x-variables affect this dummy y, i.e. how one option is preferred to the other. These x-variables are weekly hours of exercise (phys_act_h) and whether or not they presently are fitness club members (fitness_c; yes = 1, no = 0). We want to know, without formalizing any hypotheses, how more exercising per week and being a fitness center member (as opposed to not) affect the *probability* of preferring a fixed payment structure. The linear regression /linear probability model (LPM) in [Stata-output 7.2](#) sheds light on this.

The exercise effect suggests that exercising for, say, six hours per week entails a 2.7 percentage points larger probability of preferring a fixed fee than exercising for five hours. Similarly, present fitness center members have an 18.6 percentage points larger probability of preferring a fixed fee than non-members. Both effects are significant and controlled for each other.

The LPM has intuitive appeal, as it requires nothing we haven't already learned. But it also has some potential problems. The first is that the data points will not lie close to the linear regression line (i.e. poor model fit). The second is that predictions from the LPM might be smaller than 0 or larger than 1, something which is impossible for a proportion. That is, the probability of something occurring (like preferring fixed pay) must always be in the 0–1 region, just as a proportion is always between 0 and 100%. The third and minor problem is that the error term, e , always will be heteroscedastic and non-normally distributed. We'll get back to whether or not these issues are serious enough to abandon the LPM altogether in [section 7.4](#).

7.3 The standard multiple logistic (logit) regression model

Before going into the technicalities of logistic regression analysis, [Stata-output 7.3](#) first presents the logistic (logit) regression command and the results for the data analyzed in [Stata-output 7.2](#).

Two features are of note in the output. First, both independent variables have significant effects on the pay preference dummy; z-values correspond roughly to t-values.³ Second, the *signs* of the two logistic regression coefficients (+) tell us that students exercising a lot have a larger probability of preferring a fixed fee than those exercising little, and that current fitness center members have a larger probability of preferring a fixed fee than non-members. Both findings are in keeping with the results of the LPM in [Stata-output 7.2](#). But other than this, the logistic coefficients unfortunately do not tell

us much about the magnitude of these effects – as opposed to the similar effects in the LPM. Why? The answer is to be found in the upcoming mathematical properties of the logistic (logit) regression model.

The statistical model actually estimated in logistic (logit) regression is

$$\ln(p/(1-p)) = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n,$$

where $\ln$ means “the logarithm of”, and p is the probability of $y = 1$ (here: pay preference = fixed fee). That is, the left-hand side of the equation is a *non-linear* function of the independent variables. This entails that the logistic regression line is not linear but sigmoid/S-shaped, as illustrated in [Figure 7.1](#) in the case with one independent variable, x_1 . The non-linear transformation thus solves the model fit problem (partly) and the prediction problem, which are now in the 0–1 region. But this rescue operation comes at a price we have seen: non-informative coefficients, i.e. coefficients *not* having the “if x goes up with one unit, y changes with... interpretation”. On the contrary, the figure shows that the effect of x_1 – i.e. “if x_1 goes up with one unit, y changes with...” – is *not* constant, but contingent on where on the x -axis the increase in x_1 takes place.

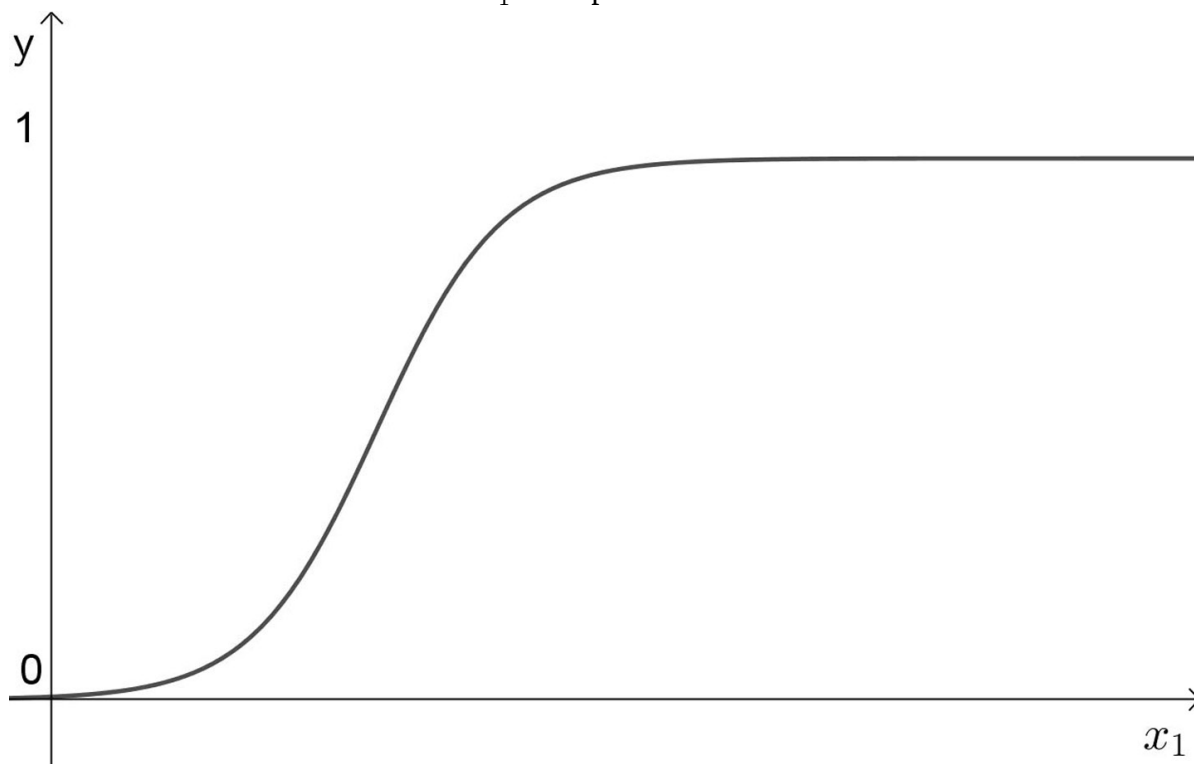


Figure 7.1 A sigmoid relationship between x_1 and y .

7.4 Interpreting and presenting the results of the logistic (logit) regression model

The non-linearity embedded in the mathematical properties of the logistic regression model is the reason why logistic regression coefficients have no translucent interpretation. And for this reason we have to make them interpretable to ourselves and to our readers. Several possibilities exist in this regard, and I present two of them: marginal effects and predicted probabilities.⁴

Marginal effects. The marginal effect approach, to simplify, focuses on the area in [Figure 7.1](#) where the S-shaped logistic regression line is very close to its linear counterpart, i.e. in the middle region of

the x -axis. In this region the marginal effect has the linear-approximation interpretation “change in y in percentage points given a one-unit increase in x_1 ”. After the logistic regression command in [Stata-output 7.3](#), one writes the command asking for marginal effects. [Stata-output 7.4](#) repeats the logit regression in [Stata-output 7.3](#) and shows the command for obtaining the marginal effects (“margins, dydx(*)”) and the results.

The marginal effects appear in the “dy/dx” column.⁵ Students exercising for, say, six hours per week have a 3.2 percentage points larger probability of preferring fixed pay than students exercising for five hours a week. (Two more hours = 6.4 percentage points and so on.) Also, present fitness center members have a 17.7 percentage points larger probability of preferring fixed pay than non-members. The “similar” marginal effects for the LPM were 2.7 and 18.6 (see [Stata-output 7.2](#)). Both marginal effects are controlled for each other as in the multiple linear regression case.⁶

The linear probability model (LPM) versus logistic regression. [Section 7.2](#) mentioned that the linear regression/the LPM was not ideal for examining a binary y . In the present case, however, the results of the LPM and the logistic regression gave qualitatively similar results. This also seems to be the *general* case. For this reason, it has become customary, especially among economists (see Angrist and Pischke, 2009), to actually prefer the LPM when y is binary. My personal take is also that the very translucent linear regression model/LPM works fine in most cases, especially when the proportion of $y = 1$ lies in the 20 to 80% region.

The marginal effect approach’s merit is ease of interpretation and communication; it is a single and transparent figure. On the downside, the base probability – that 73% in our case prefer the fixed pay form – often gets lost along the way. This problem is solved in the predicted probabilities approach coming up next.

Predicted probabilities. The name of this approach says it all. Plug different values for the x -variables into the logistic regression equation and calculate the predicted probability of $y = 1$. In this regard we should use theoretically relevant and/or representative x -values to construct probabilities. What is the predicted probability of preferring fixed pay for a student exercising four hours per week who at present is not a fitness center member? We get the answer by plugging the respective values (i.e. 4 and 0) into the equation (see Menard, 1995, p. 13)

$$e^{b_0 + b_1 x_1 + b_2 x_2} = e^{b_0 + b_1 x_1 + b_2 x_2}.$$

This yields (see the constant and the logistic regression coefficients in Stata-outputs 7.3 or 7.4)

$$e^{-0.1939 + 0.1818(4) + 0.9831(0)} = e^{-0.1939 + 0.1818(4) + 0.9831(0)} = e^{0.53331} = e^{0.5333} = 1.70452. 1.7045 = 0.6302 \approx 63\%.$$

7

A student presently not member of a fitness center who exercises for four hours per week has a 63% probability of preferring fixed pay. What if the “same” student is a fitness center member? By replacing 0 with 1 in the equation above (i.e. 0.1818×1), we get $\approx 82\%$. The otherwise similar student has an 82% probability of preferring a fixed pay form, i.e. a 19 percentage points larger probability. To get the predicted probability for students exercising very much, say 12 hours per week, we replace 4 with 14 in the equation and so on.

To do the calculations above you don't need a statistics program. But Excel might help. In Stata these calculations may be done semi-automatically. [Stata-output 7.5](#) illustrates for four hours of exercise and not being a fitness center member. (Immediately after the logistic (logit) regression you write the command “`margins, at(phys_act_h=4 fitness_c=0)`”.)

As we see, Stata yields the same result (i.e. 0.6302) as did our hand calculation above.

In most practical applications of logistic regression there are several independent variables and not only two. This makes computing of predicted probabilities more cumbersome by hand, but the principle shown above still applies. Also, since only one or two independent variables are of main interest in most settings, control variables may often be set to their mean/median values. A possible exception is dummy controls, where means often are meaningless.⁸ Predicted probabilities for different values of a numerical or ordinal x-variable may also be put into an illustrative graph.

7.5 Interaction models, non-linearity and model fit

Interaction models. Interaction (moderator) logistic regression models are estimated and interpreted in the same manner as in the linear regression case (see [section 6.2](#)). That is, one adds an interaction term to the additive model. If this term has a significant effect, the interaction/multiplicative model is preferred to the additive model. In contrast, if the interaction term is insignificant, the additive model is kept. We continue with the pay preference case from [sections 7.2/7.3](#), but the independent variables are now hours of exercise and gender (1 = women, 0 = men). The product of these two independent variables is first generated and then added as a third independent variable to the logistic regression model; [Stata-output 7.6](#) illustrates.

Since the logistic regression coefficients have no intuitive metric, the main concern is the sign and significance level of the interaction term. The fact that it is significant ($p < 0.05$) tentatively suggests that the effect of hours of exercise on preference for a fixed pay structure is *different* for men and women.⁹ The sign of the interaction term womphys (+) indicates that the exercise effect is stronger (0.321; $0.178 + 0.143 = 0.321$) for women than for men (0.178). A graph aids interpretation; see [Figure 7.2](#). Note the somewhat steeper regression line for women (black line). Whether or not this gender difference is large enough to warrant attention is another matter, however. Note also that the two non-linear regression lines are a direct consequence of the inherent non-linearity in the logistic regression model. This does *not* imply that these relationships are non-linear in real life. To examine the latter, we must address non-linearity explicitly. This is the next topic coming up.

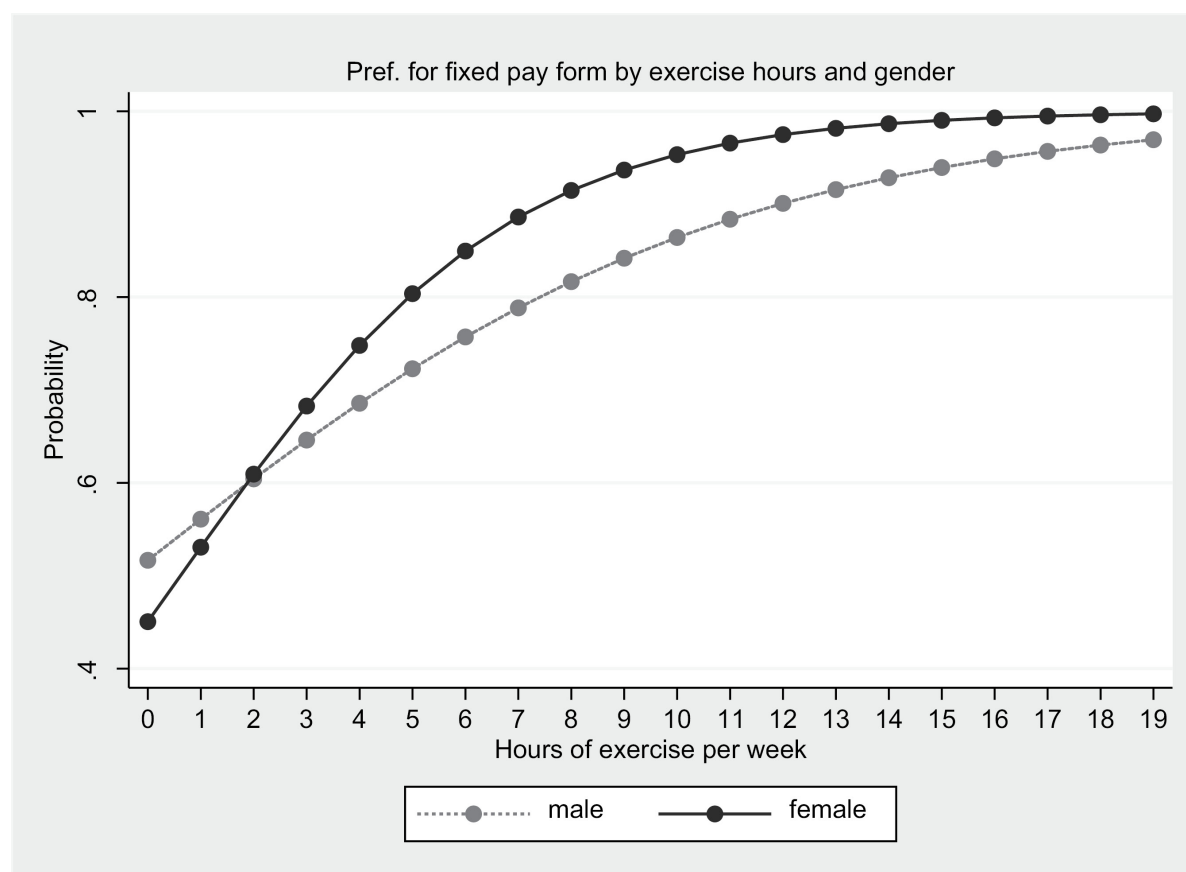


Figure 7.2 Visual illustration of interaction effect between gender and hours of exercise per week, based on the logistic (logit) regression model reported in [Stata-output 7.6](#).

Non-linearity. In the same vein as for linear regression models, the relationship between x_1 and a dummy y may be non-linear in real life. Two of three procedures illustrated in [Chapter 6](#) are relevant in the logistic case: quadratic models and sets of dummy variables. (The use of logarithms is not viable, since this feature is already embedded in the logistic model.) The procedures are exactly the same in the linear and the logistic case, but one should in the latter case also transform the logistic coefficients into marginal effects and/or predicted probabilities or graphs to get them interpretable.

Model fit. Model fit could be relevant for logistic regression models, but not so much as in the linear regression case. The reason is that there is no universal measure of model fit, such as R^2 . In contrast, there are several pseudo R^2 measures available. The default in Stata is McFadden's pseudo R^2 ; see [Stata-output 7.6](#). This roughly suggests that the independent variables explain 8% of the “variation” in pay preference form. Using the command “fitstat” immediately after the logistic regression model yields a suite of different pseudo R^2 measures, as illustrated in [Stata-output 7.7](#). The pseudo R^2 values range from 3 to 17% in the present case.

7.6 Extension: When y has three or more unordered alternatives (multinomial logistic regression)

Sometimes choice variables have more than two *unordered* alternatives or options, such as choosing between three president candidates or between walking, cycling or going by car to work/campus. In the same fashion as above, independent variables may affect how one alternative is preferred to the

other two. We continue with the data on payment preference for exercising in a fitness center, which in [sections 7.2 and 7.3](#) was simplified to two payment options. In reality there were three: (i) a fixed monthly fee, (ii) pay per time and (ii) a mixed payment scheme, i.e. a smaller fixed fee than in (i) and a larger cost per time than in (ii). In the analyses above, options (ii) and (iii) were collapsed into one alternative. The distribution for the original payment preference variable appears in [Stata-output 7.8](#).

As before, 73% of the students prefer the fixed payment scheme. The new information is how those *not* preferring fixed pay (i.e. 27%) are divided among paying per time exclusively or preferring a mixed payment scheme. The former is most popular with almost 18% as opposed to just under 10%. The question now is how hours of weekly exercise, present fitness center membership and gender affect the choice among the three payment options. When a choice variable has three or more alternatives or options, we prefer the multinomial logistic regression model and not the “ordinary” logistic.¹⁰ The command and the results are displayed in [Stata-output 7.9](#).

Interpretation of multinomial logistic models is not easy to begin with. The trick is to note that one of the options serves as baseline in all coefficient assessments. In the analysis above, this reference (i.e. base outcome) is the exclusive pay per time option. This means that paying a fixed fee per month (lower part of the output) or preferring a mixed payment scheme (upper part of the output) should *both be contrasted* with the pay per time option.¹¹ Starting with the lower part of the output, we see that students exercising a lot have a larger probability of preferring a monthly fixed fee rather than paying per time than those exercising little (i.e. a positive and significant coefficient). Similarly, present fitness center members have a larger probability than non-members of preferring a monthly fixed fee rather than paying per time (i.e. a positive and significant coefficient). In contrast, there is no gender difference in this regard since the gender coefficient is insignificant. Turning to the upper part of the output, we have that students exercising a lot are close to having a larger probability of preferring a mixed pay scheme rather than paying per time than those who exercise little (i.e. a positive and almost significant coefficient; $p = 0.056$). We also note that present fitness center members have a larger probability of preferring a mixed pay option rather than paying per time than non-members (i.e. a positive and significant coefficient). Finally, women are less inclined than men to prefer a mixed pay scheme as opposed to paying per time (i.e. a negative and significant coefficient).

How much larger/smaller probabilities are we talking about? As for binary logistic (logit) regression, multinomial logistic regression coefficients have no translucent metric. In order to shed light on the “how much” questions, therefore, the coefficients must be transformed into marginal effects or predicted probabilities for observations with different values on the x-variables in the same way as illustrated in [section 7.4](#).

The above analysis had three options/alternatives for y . The extension to models with four or more options is straightforward. Interaction effects and non-linearity must be examined in the same way as in the binary y case.

7.7 Other related issues and further reading

More can be said about regression models for non-numerical y -variables, and some of it will be covered in [Chapter 8](#). I close this chapter with some issues that often come up when analyzing binary y -variables and some recommendations for further reading.

Logit versus probit models. In certain disciplines, especially economics, a probit model is more in vogue than the logit model considered here. The difference between these models boils down to

mathematical technicalities that have no bearing on the substantive results. In Stata, probit models are estimated with the command “probit y x1 x2”. A rule of thumb says that logit coefficients often are about 1.7 times larger than probit coefficients (Long and Freese, 2014). A third alternative is the so-called *scobit* (see Nagler, 1994).

Sequential choices. Some choices are sequential. This means that one first chooses between option A or B and then, conditional on choosing A, chooses between alternative (i) or (ii). In contrast, if one chose B first, the next choice could be a no-choice or the choice between (iii) or (iv). Two examples: (a) You first decide to go on vacation (yes/no) and, once that decision is made, you choose between, say, travelling abroad or travelling domestic. If you first decided no, the second choice is a no-choice. (b) You first decide between public or private transportation and, given the public choice, choose between bus or train. If you choose private first, the second choice could be between a car or a bicycle or a motorcycle. The point from a statistical point of view is that the second choice is made by fewer people than the first, and this makes model estimation more complicated. Train (2009) has more to offer on this.¹²

The assumptions of logistic (logit) regression analysis. The assumptions for linear regression covered in Chapter 5 do in the main also apply for logistic (logit) regression analysis. Note, however, that heteroscedasticity is an even more severe problem in the logistic case.

Mediation models in logistic regression. Mediation may also come into play in logistic regression settings, but unfortunately they are not so straightforward to estimate as in the linear regression case. Breen et al. (2013) is the starting point for the study of mediation in logistic regression analysis.

Further reading

Long and Freese (2014) and Train (2009) are already mentioned in this regard. Other works on how to estimate regression models for categorical/non-numerical y-variables include Long (1997), Hosmer et al. (2013) and Best and Wolf (2015).

Notes

- 1 Binary logistic (logit) regression models are often couched in terms of voluntary choices between A and B. However, since the voluntariness in many of these choices could be questioned, quotation marks are sometimes in order. Long (1997) offers more on the various theoretical motivations for regression models on categorical y-variables.
- 2 Some questions are direct yes or no questions, i.e. 1 versus 0. Others may be recoded into one, such as voting democrat (1) or republican (0) and travelling abroad during vacation (1) or travelling domestic (0). What we label 1 and 0 depends on the research question at hand, but it is most common to denote yes as 1 and no as 0.
- 3 In linear regression we are concerned about what happens to the conditional mean of y when x_1 (or x_2 or ...) increases by one unit. This calls for t -values. In logistic regression we similarly examine what happens to the conditional *proportion* for which $y = 1$. This, without going into details, calls for z -values. As regards significance levels (p), this amounts to the same.
- 4 A third way to describe how x_1 affects y in logistic regression, popular in some areas of research, is by means of so-called odds ratios. I am no advocate for this approach since I find the terms “odds” and “odds divided by odds” (i.e. odds ratios) to be everything but transparent. I am not alone in this assessment; see for example Long and Freese (2014).

- 5 The present marginal effect is the “average marginal effect” (AME). There are other kinds of marginal effects, such as the marginal effect at the mean of all (control) variables (MEM). See Long and Freese (2014) for more details on this.
- 6 The marginal effects at the mean (MEM; cf. note 5) are very close to the AME-effects reported in the main text: 0.0327 (hours of exercise) and 0.1791 (fitness center membership).
- 7 Since the natural logarithm (i.e. Ln on a calculator) is used on the left-hand side of the logistic regression, we must use the antilogarithm (i.e. e^x on a calculator) to back-transform into probabilities (e.g. $\text{Ln } 5 = 1.6094$; $e^{1.6094} = 5$).
- 8 When, say, gender is coded 1 for women and 0 for men, the mean of gender becomes 0.4 when there are 40% women in the data. Yet in real life students are either women (1) or men (0), so setting gender to 0.4 makes little sense. An option more in sync with reality is setting gender to either 1 (women) or 0 (men). This choice, however, might affect the *magnitude* of the effects of the other independent variables in the model, due to the non-linearity inherent in the logistic model (Long and Freese, 2014). This is *not* the same as an interaction effect, which must be examined in the usual way by adding an interaction term to the model (see [section 7.5](#)). The same reasoning applies for other dummy variables.
- 9 Some caution is required when making this inference; see Ai and Norton (2003) for a strategy to become more sure in this regard.
- 10 The technicalities are covered in detail by Long (1997) and by Train (2009).
- 11 The command “base(0)” ensures this, whereas “base(2)” and “base(1)” will, respectively, put the fixed fee and the mixed pay form in the reference group. This changes the values of the coefficients but not how the overall results of the model should be interpreted.
- 12 More complicated models include aspects of the alternatives as independent variables, such as for example the price or travel time of the different alternatives in the transportation example. Again, see Train (2009) for more on this.

Exercises with data and solutions

Exercise 7A

Use the data `stud_tourism.dta`. A dummy in these data is called “more_angst” and refers to whether or not a student answers yes (1) or no (0) to the question “Due to the fear of terror attacks, are you more anxious than before about traveling to a European big city?”. Another dummy is gender, coded 1 for females and 0 for males. The third variable of interest, `num_trip`, is the answer to the question “How many trips have you done to big cities in Europe during the last two years?”.

content-type="HeadParaGroup">7A1: Estimate the following logit regression model:

$$\text{more_angst} = \text{gender} + \text{num_trip}$$

What does this multiple logistic regression tell you?

7A2: What is the predicted probability of being more anxious about traveling than before for a female student having taken three trips to big cities in Europe during the last two years? And what is the “similar” probability for a male student?

7A3: What is the predicted probability of being more anxious about traveling than before for a female student having taken one trip to big cities in Europe during the last two years? And what is the “similar” probability for a female student having taken five trips?

7A4: Is there an interaction effect between gender and `num_trip`, indicating that travel experience affects `more_angst` differently for males and females?

Exercise 7A, solution

7A1:

The logit regression coefficient for gender is 0.5597, suggesting that female students are significantly *more* likely (p value is 0.007) to report being more anxious about traveling than male ones. The logit regression coefficient for num_trip is -0.1175 , implying that students with more travel experience are significantly *less* likely (p value is 0.003) to report being more anxious about traveling than students with less travel experience. However, these coefficients tell us nothing about the magnitude of these effects.

7A2:

The predicted probability for females is 38.5%; the similar predicted probability for males is 26.4%. In other words, females have a 12 percentage points larger probability of being more anxious than “similar” males with respect to travel experience.

7A3:

The first predicted probability is about 44%; the second is about 33%. More travel experience makes female students less anxious about traveling (dynamically interpreted).

7A4:

Since the coefficient for interaction term is small (-0.0120) and insignificant (p value is 0.889), the answer is no.

```
. use "C:\Users\700245\Desktop\temp_myd\pay_pref_fit.dta", clear
```

```
. tab fixed_pm
```

fixed_pm	Freq.	Percent	Cum.
no	181	27.38	27.38
yes	480	72.62	100.00
Total	661	100.00	

```
. sum fixed_pm
```

Variable	Obs	Mean	Std. Dev.	Min	Max
fixed_pm	661	.7261725	.4462592	0	1

ta-output 7.1 Descriptive statistics for the dummy variable payment preference (fixed_pm).

```
. reg fixed_pm phys_act_h i.fitness_c
```

Source	SS	df	MS	Number of obs	=	661
Model	14.5599182	2	7.27995912	F(2, 658)	=	40.98
Residual	116.877298	658	.177625073	Prob > F	=	0.0000
				R-squared	=	0.1108
				Adj R-squared	=	0.1081
Total	131.437216	660	.199147297	Root MSE	=	.42146

fixed_pm	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
phys_act_h	.0268293	.0055132	4.87	0.000	.0160037	.0376548
fitness_c						
yes	.1863895	.0353186	5.28	0.000	.1170388	.2557403
_cons	.5078247	.0300811	16.88	0.000	.4487583	.5668912

ta-output 7.2 Linear regression/LPM of payment preference (fixed_pm) on weekly hours of exercise (phys_act_h) and fitness center membership (fitness_c).

```
. logit fixed_pm phys_act_h i.fitness_c
```

```
Iteration 0: log likelihood = -388.026
Iteration 1: log likelihood = -349.56718
Iteration 2: log likelihood = -347.82746
Iteration 3: log likelihood = -347.82093
Iteration 4: log likelihood = -347.82093
```

Logistic regression	Number of obs	=	661
	LR chi2(2)	=	80.41
	Prob > chi2	=	0.0000
Log likelihood = -347.82093	Pseudo R2	=	0.1036

fixed_pm	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
phys_act_h	.1817575	.037516	4.84	0.000	.1082274	.2552875
fitness_c						
yes	.9830869	.2037055	4.83	0.000	.5838314	1.382342
_cons	-.1938905	.1652578	-1.17	0.241	-.5177898	.1300088

ta-output 7.3 Logistic regression of payment preference (fixed_pm) on weekly hours of exercise (phys_act_h) and fitness center membership (fitness_c).

```
. logit fixed_pm phys_act_h i.fitness_c
```

Output omitted ...

```
Logistic regression                                Number of obs    =          661
                                                    LR chi2(2)       =          80.41
                                                    Prob > chi2      =          0.0000
Log likelihood = -347.82093                        Pseudo R2       =          0.1036
```

fixed_pm	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
phys_act_h	.1817575	.037516	4.84	0.000	.1082274	.2552875
fitness_c						
yes	.9830869	.2037055	4.83	0.000	.5838314	1.382342
_cons	-.1938905	.1652578	-1.17	0.241	-.5177898	.1300088

```
. margins, dydx(*)
```

```
Average marginal effects                        Number of obs    =          661
Model VCE      : OIM
```

```
Expression      : Pr(fixed_pm), predict()
dy/dx w.r.t.    : phys_act_h 1.fitness_c
```

		Delta-method				
	dy/dx	Std. Err.	z	P> z	[95% Conf. Interval]	
phys_act_h	.031917	.0062164	5.13	0.000	.0197331	.0441009
fitness_c						
yes	.1772555	.0357648	4.96	0.000	.1071577	.2473533

Note: dy/dx for factor levels is the discrete change from the base level.

ta-output 7.4 Logistic regression of payment preference (fixed_pm) on weekly hours of exercise (phys_act_h) and fitness center membership (fitness_c), with marginal effects.

```
. margins, at(phys_act_h=4 fitness_c=0)
```

```
Adjusted predictions                        Number of obs    =          661
Model VCE      : OIM
```

```
Expression      : Pr(fixed_pm), predict()
at              : phys_act_h      =          4
                  fitness_c      =          0
```

		Delta-method				
	Margin	Std. Err.	z	P> z	[95% Conf. Interval]	
_cons	.630215	.0277869	22.68	0.000	.5757537	.6846763

ta-output 7.5 Predicted probability of y = 1 for selected values of the independent variables.

```

. gen womphys = phys_act_h * gender

. logit fixed_pm phys_act_h gender womphys

Iteration 0:    log likelihood =   -388.026
Iteration 1:    log likelihood = -358.08102
Iteration 2:    log likelihood = -356.88998
Iteration 3:    log likelihood = -356.88651
Iteration 4:    log likelihood = -356.88651

Logistic regression                                Number of obs      =          661
                                                    LR chi2(3)         =          62.28
                                                    Prob > chi2        =          0.0000
Log likelihood = -356.88651                        Pseudo R2         =          0.0803

```

	fixed_pm	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
phys_act_h		.1783637	.0496193	3.59	0.000	.0811116 .2756157
gender		-.2649659	.345209	-0.77	0.443	-.9415631 .4116312
womphys		.1431828	.0721017	1.99	0.047	.001866 .2844995
_cons		.0666088	.2790297	0.24	0.811	-.4802794 .6134969

ta-output 7.6 Logistic regression of payment preference (fixed_pm) on weekly hours of exercise (phys_act_h) and gender (gender) and their product variable (womphys).

```
. fitstat
```

	logit
Log-likelihood	
Model	-356.887
Intercept-only	-388.026
Chi-square	
Deviance (df=657)	713.773
LR (df=3)	62.279
p-value	0.000
R2	
McFadden	0.080
McFadden (adjusted)	0.070
McKelvey & Zavoina	0.170
Cox-Snell/ML	0.090
Cragg-Uhler/Nagelkerke	0.130
Efron	0.095
Tjur's D	0.093
Count	0.735
Count (adjusted)	0.033
IC	
AIC	721.773
AIC divided by N	1.092
BIC (df=4)	739.748
Variance of	
e	3.290
y-star	3.964

ta-output 7.7 Different measures of model fit (pseudo R^2 measures) for the logistic regression model in Stata-output 7.6.

```
. tab payment_pref
```

payment_pref	Freq.	Percent	Cum.
pay per time	118	17.85	17.85
mixed	63	9.53	27.38
fixed per month	480	72.62	100.00
Total	661	100.00	

ta-output 7.8 Descriptive statistics for the variable payment preference (payment_pref).

```
. mlogit payment_pref phys_act_h i.fitness_c i.gender, base(0)
```

```
Iteration 0: log likelihood = -504.99568
Iteration 1: log likelihood = -457.70119
Iteration 2: log likelihood = -453.37223
Iteration 3: log likelihood = -453.33405
Iteration 4: log likelihood = -453.33403
```

Multinomial logistic regression	Number of obs	=	661
	LR chi2(6)	=	103.32
	Prob > chi2	=	0.0000
Log likelihood = -453.33403	Pseudo R2	=	0.1023

payment_pref	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
pay_per_time	(base outcome)					
mixed						
phys_act_h	.1240418	.0649641	1.91	0.056	-.0032854	.251369
fitness_c						
yes	.8417288	.367798	2.29	0.022	.120858	1.5626
gender						
female	-.8424532	.3355947	-2.51	0.012	-1.500207	-.1846996
_cons	-.7616118	.3593536	-2.12	0.034	-1.465932	-.0572918
fixed_per_month						
phys_act_h	.2446246	.0485079	5.04	0.000	.149551	.3396983
fitness_c						
yes	1.314994	.2644774	4.97	0.000	.7966278	1.83336
gender						
female	-.1018814	.2497513	-0.41	0.683	-.591385	.3876222
_cons	.0339501	.2729352	0.12	0.901	-.5009931	.5688933

ta-output 7.9 Multinomial logistic regression of payment preference (payment_pref) on weekly hours of exercise (phys_act_h), fitness center membership (fitness_c) and gender (gender).